

- ### SECTION-D
- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the purposes and necessity of providing the “Through Stones” in stone masonry walls with the help of a diagram.
- Q.37 Describe the various causes of earthquakes.
- Q.38 With the help of diagram, explain the necessity of providing the seismic bands in a masonry building constructions to make it an earthquake resistant building.

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- Q.6 _____ is an essential attribute of an earthquake resistant design of a structure that serves as a shock absorber in a structure and reduces the transmitted to one that is sustainable.
- a) Elasticity b) Ductility
c) Plasticity d) All of the above
- Q.7 Which seismic waves cause the most damage during an earthquake?
- a) P-waves b) S-waves
c) Surface waves d) Body waves
- Q.8 _____ analysis is used to obtain design seismic force.
- a) Elastic Analysis b) Plastic Analysis
c) Dynamic Analysis d) Both (a) and (b)
- Q.9 The epicentre of an earthquake is the _____
- a) Point where P-waves touch the surface
b) Nearest point to the surface from the focus perpendicular to the earth's surface.
c) Point of intersection of P and s wave fronts.
d) Antipodal point.
- Q.10 The National Institute of Disaster Management (NIDM) was established on _____
- a) September 23, 2002 b) August 14, 2001
c) October 16, 2003 d) August 14, 2004

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The longest phase of disaster management is _____ (Mitigation/ Recovery)
- Q.12 Fibre Reinforced Plastics (FRP) have _____ tensile strength and _____ weight. (High, Low/ Low, High)
- Q.13 Inertia forces in buildings act in the _____ direction to earthquake forces. (Opposite/Same)

- Q.14 _____ of steel due to environmental attack, can result in reduction of area of steel and further load carrying capacity. (Corrosion/ Carbonation)
- Q.15 More the flexibility of a building _____ is the value of fundamental period 'T' (Lower/Higher)
- Q.16 The waves which can pass through solids but fail to propagate through fluids are _____ (S-waves/ P-waves)
- Q.17 The rectangular building suffers _____ damage during an earthquake as compared with square buildings. (More/Less)
- Q.18 As per IS 1893:2002 India is divided into _____ seismic zones. (Five/Four)
- Q.19 In _____ method, the structure and foundation are separated by bearing or flexible layer. (Base isolation /Shear wall)
- Q.20 Protecting and promoting human rights is a core objective of _____ in social development.

SECTION-C

- Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)
- Q.21 Describe the "Out-of Plane Failure" as a mode of failure during an earthquake.
- Q.22 Write a short note on "Psychology of Rescue".
- Q.23 Describe the various grades of concrete recommended by IS: 13920-1993 to be provided for construction of building to make them as earthquake resistant buildings.
- Q.24 Enlist the various rescue equipment required to carry out the rescue operations, for damage control and casualty management in a residential area consist of high rise buildings.
- Q.25 Describe the various effects on seismic performance of a building if it is "Irregular in Plan or Shape".
- Q.26 Describe the procedure of repair of cracks developed due to an earthquake in a residential masonry building.